

Question:

An organization is maintaining the data of employee according to cadre of employee with following parameters: name, id, designation, salary, promotion status. Apply the constructor overloading to implement it.

Steps:

1. Create an `Employee` class.
2. Declare variables for employee name, ID, designation, salary, and promotion status.
3. Create a constructor with `name` and `id`.
4. Assign default values for designation, salary, and promotion status.
5. Create a second constructor with `name`, `id`, `designation`, and `salary`.
6. Assign the given values and set promotion status as `"Not Promoted"`.
7. Create a third constructor with all five employee details.
8. Assign all the given values.
9. Create a `display()` method to display employee details.
10. Create the `AllEmployees` class with the `main()` method.

11. Create three employee objects using the three different constructors.
12. Call the `display()` method for each employee.
13. Display all employee details.
14. End the program.

Algorithm:

1. **Start**
2. Define the `Employee` class with employee details.
3. Create a constructor with `name` and `id`.
4. Set default designation as "`Employee`", salary as `20000`, and promotion status as "`Not Promoted`".
5. Create a second constructor that accepts `name`, `id`, `designation`, and `salary`.
6. Set promotion status as "`Not Promoted`".
7. Create a third constructor that accepts all employee details.
8. Initialize all the employee details.
9. Create a `display()` method to print the employee information.
10. In the `main()` method, create three employee objects using different constructors.
11. Call `display()` for each employee.
12. **Stop.**

```
File Edit Selection View Go Run Terminal Help
J Bank.java 1 J CalculateMarks.java 1 J Palindrome.java 1 J AllEmployees.java 1 X
C:\Users\venik\Documents\Records\week4,5 > J AllEmployees.java > {} week4
1 package week4;
2
3 class Employee {
4
5     String name;
6     int id;
}

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\venik> & 'C:\Program Files\Java\jdk-25.0.3\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\venik\va-project\bin' 'week4.AllEmployees'
Name: Ravi
ID: 101
Designation: Employee
Salary: 20000.0
Promotion Status: Not Promoted

Name: Anu
ID: 102
Designation: Manager
Salary: 50000.0
Promotion Status: Not Promoted

Name: Kiran
ID: 103
Designation: Developer
Salary: 45000.0
Promotion Status: Promoted

PS C:\Users\venik>
```

Question:

Design a “farm animals” java application with the details of animals like cow, pig, horse. Consider the following details like where they stay, what they eat, the sound they make by using classes and objects.

Steps:

1. Create a class named `FarmAnimal`.
2. Create a `display()` method inside the class.
3. Display the details of the cow, including its place of stay, food, and sound.
4. Display the details of the pig, including its place of stay, food, and sound.
5. Display the details of the horse, including its place of stay, food, and sound.
6. Create a `Main` class containing the `main()` method.
7. Create an object of the `FarmAnimal` class.
8. Call the `display()` method using the object.
9. Display all animal details.
10. End the program.

Algorithm

1. **Start**
2. Define the `FarmAnimal` class.
3. Define the `display()` method.
4. Display cow details:
 - Stays in Cowshed
 - Eats Grass
 - Sound: Moo
5. Display pig details:
 - Stays in Pigsty
 - Eats Corn
 - Sound: Oink

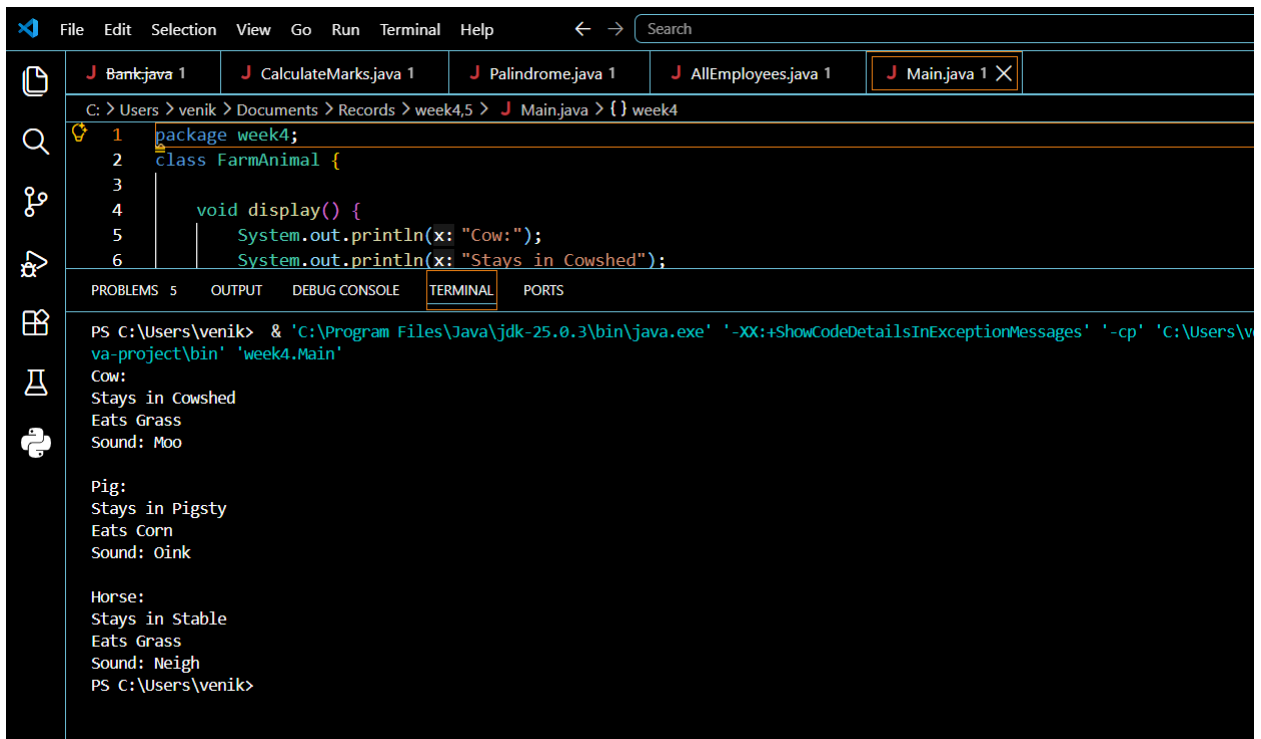
6. Display horse details:

- Stays in Stable
- Eats Grass
- Sound: Neigh

7. Create an object of the `FarmAnimal` class.

8. Call the `display()` method.

9. **Stop.**



The screenshot shows an IDE with a dark theme. The top menu bar includes File, Edit, Selection, View, Go, Run, Terminal, and Help. Below the menu is a search bar. The editor has several tabs: Bank.java 1, CalculateMarks.java 1, Palindrome.java 1, AllEmployees.java 1, and Main.java 1 (which is active and highlighted with a red border). The active tab shows the following code:

```
1 package week4;
2 class FarmAnimal {
3
4     void display() {
5         System.out.println(x: "Cow:");
6         System.out.println(x: "Stays in Cowshed");
7     }
8 }
```

Below the editor, there are tabs for PROBLEMS (5), OUTPUT, DEBUG CONSOLE, TERMINAL (which is active and highlighted with a red border), and PORTS. The TERMINAL tab shows the following output:

```
PS C:\Users\venik> & 'C:\Program Files\Java\jdk-25.0.3\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\venik\va-project\bin' 'week4.Main'
Cow:
Stays in Cowshed
Eats Grass
Sound: Moo

Pig:
Stays in Pigsty
Eats Corn
Sound: Oink

Horse:
Stays in Stable
Eats Grass
Sound: Neigh
PS C:\Users\venik>
```